

**Site Acceptance Test Plan
(SATP)**

For

Plate Recognition System

**Saudi Aramco, WASEA BULK PLANT
GATE**

Prepared By



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DOCUMENT CONTROL

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1 Introduction

1.1 Scope

The Site Acceptance Test (SAT) Plan is a guide to the acceptance tests of the PRS System deployed at WASEA BULK PLANT GATE. It outlines the types of tests to be carried out, the features to be tested, pass and fail criteria, staffing matters and responsibilities, and provides a preliminary schedule for the execution of the acceptance tests.

After performing Installation, the objective of SAT is to provide NAIZAK on-site evidence that the system meets Saudi ARAMCO requirements as presented in the FSD and Equipment Submittals and can therefore be accepted to process the final handover and MCC.

1.2 Applicable Documents

- Project Specification
- Approved technical submittals

1.3 Terms and Abbreviations

Acronym	Description
ANSI	American National Standards Institute
DHTCC	Dhahran Tower Computer Centre
FAT	Factory Acceptance Test
FATP	Factory Acceptance Test Plan
FSD	Functional Specification Document
FTP	File Transfer Protocol
GUI	Graphical User Interface
IRF	Incident Report Form
LAN	Local Area Network
OCR	Optical Character Recognition
PAT	Performance Acceptance Test
PATP	Performance Acceptance Test Plan
PDR	Preliminary Design Review
PRS	Plate Recognition Systems
RSCC	Regional Security Control Center
SAES	Saudi Aramco Engineering Standards
SAT	Site Acceptance Test
SATP	Site Acceptance Test Plan
SDD	System Design Document
SDP	System Development Plan
STRR	System Readiness Review
PAT	Performance Acceptance Test

TRS	Test Record Sheet
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Term / Word	Explanation
Company	Refers to Saudi Arabian Oil Company (Saudi Aramco) organization.
Contractor	Refers to ForeVision Business Solutions
Application Software	Programs that perform specific tasks, such as word processing or database management, in contrast to system software, which runs the computer system and utilities, which help the users maintain and organize the system .
Availability	The percentage of time a system or individual module remains on line and performs as specified during observation period .
Factory Acceptance Test	The acceptance test that demonstrates compliance of the System Acceptance Test with all requirements, to the extent possible in the Contractor's Factory environment.
Site Acceptance Test	The acceptance test, performed on the installed System that demonstrates compliance with all requirements, except availability.

2 Personnel Requirements

2.1 ForeVision Personnel and Responsibilities

ForeVision's SAT Team is composed of following personnel:

Name / Position	Responsibility
DanishKhan / Project Manager	Setups SAT activities schedule and implement proper coordination with NAIZAK representatives, subcontractors and suppliers.
Zeeshan Farrukh / PRS Engineer	Conducts the SAT as per the approved SAT procedures, configuration management and to maintain the Exceptional Items list..
Gilbert & Saquib / Hardware Engineers	Handles the ANPR hardware related aspects.

2.2 NAIZAK Personnel and Responsibilities

NAIZAK Team is comprised of the following persons:

Name / Position	Responsibility
/ Test Manager	
/ Tester	
/ Test Specialist	

NAIZAK Representatives are authorized to:

- Witness all the Tests and Inspection being conducted in accordance with the Site Acceptance Test procedures.
- Raise comments or initiate corrective actions if necessary related to the conditions of the facilities, and equipment covered by the SAT safety and quality.

2.3 Authorized Signatures

All documents related to the SAT shall be signed by NAIZAK and ForeVision Representatives.

3 Management / Control

The status of the versions of hardware, software and corresponding documentation shall be managed and controlled to ensure that the correct items are tested, and keep traceability of all modifications that may occur during the SAT.

The part number or reference, serial number, version, shall be recorded in the SAT record from the beginning of the SAT until SAT completion. Any configuration change shall be recorded and commented.

Any kind of discrepancies or failures during the tests shall be noted down in the Test and Exceptional Items List Log. ForeVision shall investigate the failure after the SAT session and provide, the day after, to Saudi Aramco a detailed incident report describing the reason of the failure and the corrective action conducted. Test shall be re-conducted after approval of the Incident Report by Saudi Aramco.

3.1 Location and Date

The SAT shall take place at **WASEA BULK PLANT GATE, Saudi Aramco**

From _____ 2014

To _____ 2014

3.2 Schedule

- 09h00 – 11h30 : SAT session

3.3 Test Program

Initiation	• Team Introduction and SAT Overview
Structured Testing	• 100 – Hardware Inspection • 200 – Image Capture, Recognition and Alarm • 300 – Data Replication

4 SAT Environment

4.1 Lane

Each lane of WASEA BULK PLANT GATE shall have the following equipments:

- PRS Color Camera
- Camera Housing
- LED White Light with its Power Supply
- Rotating Warning Light (Visual Alarm)
- 12 VDC Power Supply
- All above equipment mounted over the Camera Pole
- Induction Loop installed on the lane

4.2 Gatehouse

GATE HOUSE shall have following equipments pertaining to PRS System:

- PRS Recognition Computer with ANPR License Dongle
- PRS Local Db Computer for hosting the local database
- Siren (Audio Alarm)
- Network Switch
- Relay Panel
- PCI Analog Card
- Induction Detector Device
- RS232-485 Convertor
- KVM Switch
- Monitor, Keyboard, Mouse connected with KVM Switch
- PRS Cabinet

4.3 Regional Workstation (RSCC) Ras Tanura

This Workstation is already configured & installed at RSCC Ras Tanura. The PRS Management and the Regional Monitoring Application is installed on this workstation to let the users maintain and administer the functions of PRS System for this region.

Note: Provision of RSCC Workstation is not the scope of this Project. Here this is described only to complete the test cycles.

5 Testing

For structured testing, the testers shall follow a set of documented procedures provided by ForeVision in the form of Test Sheets. The procedures contain a means to report results. Each procedure step is described with its expected result. In front of the result, two check box are provided for the testers to declare the pass or fail status of test step. To write detailed comments, the space is provided at the end of each test category.

5.1 Structured Test Outline

Following is the brief outline of Structured Tests. The testers shall use the Test Sheets designed in section 5.3 to perform the actual testing.

Test #	Function	Duration (min)
Test Group 100 – Hardware Inspection		
	Lane	60
101	PRS Color Camera	
102	Camera Housing	
103	LED White Light and its Power Supply	
104	Rotating Warning Light (Visual Alarm)	
105	12 VDC Power Supply	
106	Camera Pole	
107	Induction Loop	
	Gatehouse	
108	Cabinet	
109	Network Switch	
110	PRS Local Database Computer	
111	PRS Recognition Computer	
112	ANPR License Dongle	
113	PCI Analog Card	
114	Relay Panel	
115	Induction Detector Device	
116	RS232-485 Convertor	
117	KVM Switch	
118	Monitor, Keyboard, Mouse connected with KVM Switch	
119	Siren (Audible Alarm)	
Test Group 200 – Image Capture, Recognition and Alarm		
201	Capture Rate	15
202	Recognition Accuracy (Characters+Color)	15
203	Recognizing New and Old Saudi Plates	15
204	Recognizing Tilted and Skewed Plates	15

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205	Flagged Vehicle Alarm	15
206	Unrecognized Vehicle Alarm	15
Test Group 300 – Data replication		
301	Data transfer from Main to Local Db Computer	15
302	Data transfer from Local Db to Main Server	15
303	Corporate Network Breakdown	15

5.2 Factory Acceptance Test Report

BETWEEN

NAIZAK Global Engineering Systems
Saudi Arabia

AND

ForeVision Business Solutions
P.O.Box # 20002, Jeddah 21455, Saudi Arabia

FOR

Plate Recognition System (PRS)

NAIZAK representative hereby, confirms that Site Acceptance Test for PRS System has been completed successfully. NAIZAK further confirms that the System meets Saudi Aramco requirements as presented in the FSD, is therefore accepted for Final Handover.

Test Location:

WASEA BULK PLANT GATE, Saudi Aramco

Started On: _____

Completed On: _____

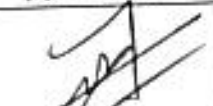
NAIZAK Representative	ForeVision Representative
Name	Name ZEESHAN
Position	Position PRS Engineer
Signature	Signature 

5.3 Test Sheets

Test Group 100 – Hardware Inspection			
Test Purpose: The purpose of this inspection is to check that PRS System hardware deployed on the site is in compliance with the specification defined in the FSD.			
Test Duration: 60 minutes		FSD Reference:	
Test Procedure: Explore the configuration of the PRS System hardware deployed on the site and check its compliance with FSD.			
Test #	Item	Observation	
	Lane		
101	PRS Color Camera	Pass ☒	Fail ☐
102	Camera Enclosure	Pass ☒	Fail ☐
103	LED White Light with its Power Supply	Pass ☒	Fail ☐
104	Rotating Warning Light (Visual Alarm)	Pass ☒	Fail ☐
105	12 VDC Power Supply	Pass ☒	Fail ☐
106	Camera Pole	Pass ☒	Fail ☐
107	Induction Loop	Pass ☒	Fail ☐
	Gatehouse		
108	Cabinet	Pass ☒	Fail ☐
109	Network Switch	Pass ☒	Fail ☐
110	PRS Local Database Computer	Pass ☒	Fail ☐
111	PRS Recognition Computer	Pass ☒	Fail ☐
112	ANPR License Dongle	Pass ☒	Fail ☐
113	PCI Analog Card	Pass ☒	Fail ☐
114	Relay Panel	Pass ☒	Fail ☐
115	Induction Detector Device	Pass ☒	Fail ☐
116	RS232-485 Convertor	Pass ☒	Fail ☐
117	KVM Switch	Pass ☒	Fail ☐
118	Monitor, Keyboard, Mouse connected with KVM Switch	Pass ☒	Fail ☐
119	Siren (Audible Alarm)	Pass ☒	Fail ☐

Test Group 100 - Hardware Inspection	
Test #	Comments

Test observed and verified by:

	ForeVision PRS Engineer	ForeVision Project Manager	Saudi Aramco Representative	NAIZAK Representative
Name	 ZEESHAN	for ZEESHAN	Yami, Salem	
Sign				

Dated: March 9, 2015

Test Group 200 - Image Capture, Recognition and Alarm

Description: The purpose of these tests is to check that image capturing, recognition and alarming features of PRS System are in compliance with the specification defined in the FSD.

Test Equipment List:

1. Several old, new, short, and long plates.
2. Tilted Plates
3. Dirty number plates
4. Hot Listed number plates

Test # 201	Capture Rate
Test Purpose: This test shall ensure that PRS System captures all vehicles entering and leaving the gates at a rate of 100%.	
Test Duration: 15 minutes	FSD Reference:
Test procedure: 1. Note down the current date & time. You shall use this date & time value in report search criteria in later steps. 2. Pass different number plates several times through the lanes and note down the count. 3. Stop the activity and note down the time. 4. Run Periodically Audit Trail Report from Admin Application by using the date & time criteria to verify the count of captures for these lanes.	
Observation:	
1	The report shows all the transactions that occurred during the experiment. Pass ☒ Fail ☐

Test # 202	Recognition Accuracy (Characters + Color)		
Test Purpose: This test shall ensure that PRS System has minimum 95% accuracy to recognize plate characters as well as color.			
Test Duration: 15 minutes		FSD Reference:	
Test procedure: 1. Note down the current date & time. You shall use this date & time value in report search criteria in later steps. 2. Pass different number plates of white, blue, red, green, yellow background colors several times through the lanes and note down the count. 3. Stop the activity and note down the time. 4. Run Recognition Summary Report from Admin Application by using the date & time criteria to estimate the recognition accuracy.			
Observation:			
1	The accuracy rate is greater than or equal to 95%.	Pass ☒	Fail ☐

Test # 203	Recognizing New and Old Saudi Plates		
Test Purpose: This test shall ensure that the PRS System is capable to recognize old (short, long) plates as well as the newly released short and long Saudi plates.			
Test Duration: 15 minutes		FSD Reference:	
Test procedure: 1. Note down the current date & time. You shall use this date & time value in report search criteria in later steps. 2. Pass different several old-short, old-long, new-short and new-long number plates through the lanes. 3. Stop the activity and note down the time. 4. Run Periodically Audit Trail Report from Admin Application by using the date & time criteria to check the results.			
Observation:			
1	All plates are captured accurately.	Pass ☒	Fail ☐

Test # 204	Recognizing Tilted Plates		
Test Purpose: This test shall ensure that PRS is capable to read any plate tilted at an angle of 15° or lesser.			
Test Duration: 15 minutes		FSD Reference:	
Test procedure: 1. Note down the current date & time. You shall use this date & time value in report search criteria in later steps. 2. Pass some plates tilted upto 15° angle through the lanes. 3. Stop the activity and note down the time. 4. Run Periodically Audit Trail Report from Admin Application by using the date & time criteria to check the results.			
Observation:			
1	All plates are recognized accurately.	Pass ☒	Fail ☐

Test # 205	Flagged Vehicle Alarm		
Test Purpose: This test shall ensure that PRS System generates the alarm for the flagged vehicles. Alarm shall be both visual and audible and shall indicate in which lane the vehicle is traveling.			
Test Duration: 15 minutes		FSD Reference:	
Test procedure: 1. Make sure that the data of flagged vehicle is recorded in the PRS System by using Admin Application. 2. Pass the flagged number plate through the lane.			
Observation:			
1	Siren is activated.	Pass ☒	Fail ☐
2	Rotating Warning Light of the respective lane is activated.	Pass ☒	Fail ☐
3	The alarm can be stopped from the Local Controller.	Pass ☒	Fail ☐

Test # 206	Unrecognized Vehicle Alarm		
Test Purpose: This test shall ensure that PRS System generates the alarm for the unrecognized vehicles. Alarm shall be both visual and audible and shall indicate in which lane the vehicle is traveling.			
Test Duration: 15 minutes		FSD Reference:	
Test procedure: 1. Pass the dirty or damaged number plate through the lane so that it cannot be recognized by the system.			
Observation:			
1	Siren is activated.	Pass ☒	Fail ☐
2	Rotating Warning Light of the respective lane is activated.	Pass ☒	Fail ☐
3	The alarm can be stopped from the Local Controller.	Pass ☒	Fail ☐

disable functions for all locations.

Test Group 200 - Image Capture, Recognition and Alarm	
Test #	Comments

Test observed and verified by:

	ForeVision PRS Engineer	ForeVision Project Manager	Saudi Aramco Representative	NAIZAK Representative
Name	ZEESHAN	for ZEESHAN	Yamir, Salem	
Sign				

Dated: March 9, 2015

Test Group 300 – Data Replication
Description: The purpose of this test is to check that speed of data replication between PRS Main server and the Local Database Computer fulfils the requirements defined in the FSD.
Test Equipment List : N/A

Test # 301	Data transfer from Main to Local Db Computer		
Test Purpose: This test shall ensure that PRS Main Server replicates the hot list data within sixty (60) seconds to the Local Database Computer.			
Test Duration: 15 minutes		FSD Reference:	
Test procedure: 1. Record a new flagged vehicle into the system by using PRS Admin Application. 2. Pass the same flagged number plate through the lane at least after 60 seconds of the record entry.			
Observation:			
1	Warning light and siren activates within 60 seconds.	Pass ☒	Fail ☐

Test # 302	Data transfer from Local Db to Main Server		
Test Purpose: This test shall ensure that Local Database Computer shall replicate its alarmed transactions within no more than one (1) second to the PRS Main Server. This test shall also ensure that the Local Database Computer shall replicate its normal (non-alarmed) transactions to the PRS Main Server within not more than thirty (30) minutes.			
Test Duration: 15 minutes		FSD Reference:	
Test procedure: 1. Pass different hot-listed and non-hot listed number plates through the lanes. 2. Run Periodically Audit Trail Report from Admin Application to see the transactions.			
Observation:			
1	The report shows the respective transactions.	Pass ☒	Fail ☐

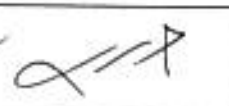
Test # 303	Corporate Network Breakdown		
Test Purpose: This test shall ensure if there is a breakdown in the communication between Local Database Computer and the Main Server, the Local Computer shall continue to store data and transfer it automatically through the database scheduled jobs after when the connection is re-established.			
Test Duration: 15 minutes		FSD Reference:	
Test procedure: 1. Break the communication between the Main Server and the Local Database Computer by un-plugging the LAN Cable. 2. Pass some number plates through the lane to generate the transactions. 3. Re-plug the LAN Cable and wait for at least 30 minutes OR the Scheduled Job manually to save the time. 4. Query the results. Now report shall show all the transactions that occurred during the network breakdown.			
Observation:			
1	Data successfully stored on the Local Db Computer during network breakdown.	Pass ☒	Fail ☐
2	Date successfully transferred to the Main Server within 30 min after re-establishment of network connection.	Pass ☒	Fail ☐

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Test Group 300 – Data Replication	
Test #	Comments

Test observed and verified by:

	ForeVision PRS Engineer	ForeVision Project Manager	Saudi Aramco Representative	NAIZAK Representative
Name	ZEESHAN	FOR ZEESHAN	Yami, Salem	
Sign				

Dated: March 9, 2015